

Erratum of The Hash Function CHIME

CHIME submission team – September 2026

Summary of changes. Items are listed against the sections of the submitted specification.

#	Location	Change
1	Table 7	The four 64-bit words of the round constant RC_i are no longer identical. The revised constants are shown in Table 1.
2	Section 1.5.2	“After absorbing the $(m-1)$ -th message block, execute $S_{m-1} \leftarrow S_{m-1} \oplus (0^{c-1} 1)$ ” is replaced by “Before absorbing the $(m-1)$ -th message block, execute $S_{m-1} \leftarrow S_{m-1} \oplus (0^{1535} 1)$ ”.
3	Section 1.5.2	“For each 448-bit message block M'_i ($i = 0, \dots, m-1$), XOR it with the first 448 bits of the current state” is replaced by “For each 448-bit message block M'_i ($i = 0, \dots, m-1$), XOR it with the 448 bits of the state $S_i[0] \dots S_i[447]$ ”.
4	Figure 3	The label IV at the capacity input is replaced by 0^c .
5	Section 5.3.2	“ $T_{512} = \max\{2^{512-\sigma+n_c}, 2^{h-\min(n_{\text{blue}}, n_{\text{red}}, n_{\text{matching}})}\}$ ” is replaced by “ $T_{512} = \max\{2^{\sigma+n_c}, 2^{h-\min(n_{\text{blue}}, n_{\text{red}}-\sigma, n_{\text{matching}})}\}$ ”.
6	Table 12	372 is replaced by 334.
7	Section 1.1	“Length of bit string output after the hash function is applied to the message” is replaced by “Bit string output after the hash function is applied to the message”.

1. Round-Constant Correction

Every component of the round function other than *AddRoundConstant* either acts independently within each 64-bit word or permutes whole words: λ is built from word-wise XORs, rotations and shifts together with word and half-block permutations, χ is bitwise, and τ only renames the state arrays. *AddRoundConstant* was therefore the only step able to distinguish the four words of a state array, and with four identical constant words it did not do so. In the submitted specification the round constant RC_i consists of four identical 64-bit words X_i (Table 7).

As a consequence the permutation P maps the set of states whose six state arrays each consist of four equal 64-bit words into itself, for any number of rounds. The set has dimension 384 bits inside a 1536-bit state.

The round constant RC_i is replaced by a constant whose four words are distinct, drawn from the same generator:

$$RC_i = X_{i,3} || X_{i,2} || X_{i,1} || X_{i,0}, \quad X_{i,k} = F(4i+k), \quad k = 0, 1, 2, 3,$$

$$F(t) = 0x\text{DEADBEEF00000000} \oplus t \oplus (0x9E3779B97F4A7C15 \times (t+1)) \bmod 2^{64}.$$

Word k of RC_i is XORed into word k of A_0 . Since the constants of the submission were $X_i = F(i)$, the generator, the golden-ratio multiplier and the injection point are the same as before. The revised round constants are given in the following Table 1 and replaces Table 7 of the submitted specification. CHIME-512 uses rows 0–14 and CHIME-1024 rows 0–19, as before.

The target security claims remain unchanged. The revised round constants do not affect the differential and linear active-S-box bounds.

2. Specification Corrections and Clarifications

Section 1.5.2. In Section 1.5.2, to make it easier for readers to understand and avoid ambiguity “For each 448-bit message block M'_i ($i = 0, \dots, m-1$), XOR it with the first 448 bits of the current state” is replaced by:

For each 448-bit message block M'_i ($i = 0, \dots, m-1$), XOR it with the 448 bits of the state $S_i[0]||\dots||S_i[447]$.

Table 1: Round Constants (Hexadecimal)

i	$X_{i,0}$	$X_{i,1}$	$X_{i,2}$	$X_{i,3}$
0	409AC7567F4A7C15	E2C34D9DFE94F82B	040BD3C37DDF743D	A670580AFD29F057
1	C9B8DE707C746C6D	6BE164B7FBBEE87B	8D29EAFD7B096495	2F167324FA53E0AF
2	515EF96A799E5CB5	F0877FD1F8E8D8DB	12CF8417783354ED	B4340A5EF77DD0F7
3	D67C908476C84D1D	79A516CBF612C92B	9BED9F31755D4535	3DDA2578F4A7C15F
4	5F02ABBE73F23D75	C14B31E5F33CB96B	60B3B62B7287359D	82F83C92F1D1B1B7
5	242142D8711C2DAD	4669CB1FF066A9DB	E85651456FB125F5	0B9ED78CEEFA1EF
6	ADC75DF26E461E15	CF0FE239ED909A3B	7174687F6CDB162D	90BCEEA6EC259257
7	32E574EC6B700E7D	542DFD53EABA8A6B	F61A03996A050695	184289C0E94F82BF
8	BB8B0E066899FE95	DDF3944DE7E47AEB	7F381AB3672EF6FD	E160A0FAE67972D7
9	00A9292065C3EF2D	A291AF67E50E6B3B	C4DE35AD6458E715	6606BA14E3A3636F
10	884FC05A62EDDF75	2BB44681E2385B5B	4DFCCCC76182D7AD	EF25550EE0CD53B7
11	116DDB746017CF9D	B35A61BBDF624BEB	D282E7E15EACC7F5	74CB6C28DDF743DF
12	9633F26E5D41C035	387878D5DC8C3C2B	5BA0811B5BD6B81D	FDE90742DB213477
13	1FD18D885A6BB06D	811E13CFD9B62C5B	234698355900A8B5	428F1E7CD84B24AF
14	E4F7A4A25795A095	063C2AE9D6E01CFB	A864B32F562A98ED	CBAD3996D57514D7
15	6D95BFDC54BF913D	8FC2C403D40A0D2B	310B4A4953548915	5373D0B0D29F057F
16	F2B856F651E98115	14E0DF3DD133FD2B	B6296563507E793D	D811EBAACFC8F5D7
17	7A5E70104F1371ED	9D86F657CE5DEDFB	3FCF7C9D4DA86995	A13782C4CCF2E5AF
18	C37C0B0A4C3D61B5	62A49171CB87DE5B	84ED17B74AD25A6D	26D59DFECA1CD677
19	480222244967521D	EA4AA86BC8B1CE2B	0DB32ED147FC4A35	AFFBB718C746C6DF

In Section 1.5.2 the lengths of S_{m-1} and of $(0^{c-1}\|1)$ are 1536 and $c = 1088$ bits, so the XOR was not defined. The constant becomes $(0^{1535}\|1)$, which is consistent with Figure 3, and the sentence is replaced by:

Before absorbing the $(m - 1)$ -th message block, execute $S_{m-1} \leftarrow S_{m-1} \oplus (0^{1535}\|1)$. Both the permutation input and the feeding-forward term of that iteration use this updated S_{m-1} .

Figure 3. The label IV at the initial capacity input is replaced by 0^c , consistent with the initialization $S_0 = 0^{1536}$.

3. Editorial Corrections

Section 5.3.2. A typographical error in the expression for T_{512} is corrected as follows:

$$T_{512} = \max \left\{ 2^{\sigma+n_c}, 2^{h-\min(n_{\text{blue}}, n_{\text{red}}-\sigma, n_{\text{matching}})} \right\}.$$

Table 12. The memory-complexity exponent 372 is corrected to 334. This was a transcription error.

Section 1.1. “Length of bit string output after the hash function is applied to the message” is replaced by “Bit string output after the hash function is applied to the message”.

Note. Big-endian in the submitted specification is replaced by little-endian, in agreement with the submitted reference and optimized implementations.

4. Package Update

The reference and optimized implementations have been updated and all known-answer test vector files regenerated for both instances.